Roll No.:

Test Date: 23-03-2025

Aakash

Medical | IIT-JEE | Foundations

E

Booklet Code

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AIATS

ALL INDIA AAKASH TEST SERIES

for

Medical Entrance Exam - 2025

National Eligibility-cum-Entrance Test (NEET)

TEST No. 7

(XII Passed Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 7

TOPICS OF THE TEST

Physics	Ray Optics and Optical Instruments, Wave optics, Dual Nature of Matter and Radiation; Atoms, Nuclei; Semiconductors, Experimental Skills
Chemistry	The *p*-Block Elements, Biomolecules, Principles Related to Practical Chemistry
Botany	Organisms and Populations, Ecosystem, Biodiversity and conservation
Zoology	Biotechnology-Principles and Processes, Biotechnology and its Applications

MM : 720

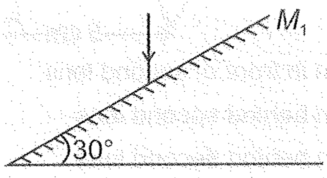
TEST - 7

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer :

1. A mirror M_1 is inclined at an angle of 30° with the horizontal as shown in figure. If a ray of light is incident vertically on mirror, then the angle of reflection will be

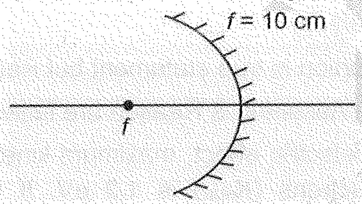


(1) 60° (2) 45°
(3) 90° (4) 30°

2. When an α-particle of mass m moving with velocity v bombards on a heavy nucleus of charge 'Ze', its distance of closest approach (r) from the nucleus depends on m as

(1) $r \propto m$ (2) $r \propto \frac{1}{m}$
(3) $r \propto \frac{1}{\sqrt{m}}$ (4) $r \propto \frac{1}{m^2}$

3. The position of object placed in front of concave mirror of focal length 10 cm so that three times magnified, virtual and erect image is formed will be

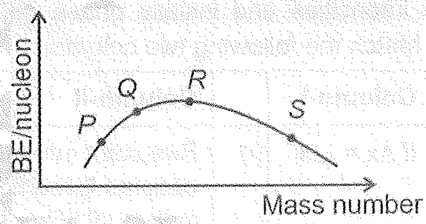


(1) – 3 cm (2) $-\frac{10}{3}$ cm
(3) $-\frac{20}{3}$ cm (4) $-\frac{20}{7}$ cm

4. For n-type semiconductors (symbols have their usual meaning)

(1) $n_e >> n_h$
(2) $n_h >> n_e$
(3) $n_h = n_e$
(4) Both (2) and (3) are correct

5. Binding energy per nucleon versus mass number curve for nuclei is shown in the figure. P, Q, R and S are four nuclei indicated on the curve. The process that would release energy is



(1) $2Q \rightarrow S$ (2) $R + 2Q \rightarrow 2S$
(3) $2R \rightarrow S$ (4) $P + Q \rightarrow R$

6. Choose the correct statement with reference to interference and diffraction.

(1) In interference and diffraction, light energy is redistributed.
(2) If light energy reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe.
(3) There is no gain or loss of energy, which is consistent with the principle of conservation of energy.
(4) All of these

Space for Rough Work

7. Light travels from a rarer medium to a denser medium, the speed and frequency of the light wave respectively
(1) Increases and decreases
(2) Decreases and increases
(3) Decreases and remains same
(4) Remains same and increases

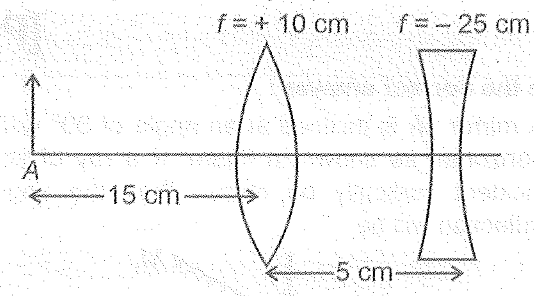
8. Monochromatic light of frequency 5×10^{14} Hz is produced by a laser. If 2×10^{16} photons are emitted by the source per second, then power emitted by the laser source is
(1) 1.16 mW (2) 3.32 mW
(3) 6.63 mW (4) 66.3 mW

9. Two waves from coherent sources meet at a point with a path difference of Δx. Both the waves have same intensities and initially phase difference is zero. Match the following two columns.

	Column-I		Column-II
a.	If $\Delta x = \lambda/3$	(i)	Resultant intensity will become three times of individual wave intensity
b.	If $\Delta x = \lambda/6$	(ii)	Resultant intensity will remain same as that of individual wave intensity
c.	If $\Delta x = \lambda/4$	(iii)	Resultant intensity will become two times of individual wave intensity
d.	If $\Delta x = \lambda/2$	(iv)	Resultant intensity will become zero

(1) a(ii), b(iii), c(iv), d(i)
(2) a(iii), b(iv), c(i), d(ii)
(3) a(iv), b(iii), c(i), d(ii)
(4) a(ii), b(i), c(iii), d(iv)

10. The position of image formed by the lens combination given below is (where f is focal length and A is position of object)



(1) 25 cm in front of second lens
(2) 25 cm behind second lens
(3) 30 cm behind second lens
(4) Infinity

11. **Assertion (A):** For a given frequency of the incident radiation, the stopping potential in photo electric effect is independent of its intensity.
Reason (R): The maximum kinetic energy of photoelectrons is independent of frequency of the light source and the emitter plate material, but is dependent on intensity of incident radiation.
(1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion
(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion & Reason are false statements

12. In photoelectric effect, maximum kinetic energy of photoelectrons (K_{max}) is 1.2 eV. If frequency is increased by 50%, then K_{max} becomes 3.6 eV. Calculate the work function of emitter plate
(1) 2.3 eV (2) 3.6 eV
(3) 4.8 eV (4) 5.6 eV

Space for Rough Work

13. As the temperature rises then viscosity for liquids and gases respectively
(1) Decreases and increases
(2) Remains same for both
(3) Increases for both
(4) Decreases for both

14. A stationary bomb suddenly explodes and splits in two unequal fragments of masses m_1 and m_2. The ratio of their de-Broglie wavelength is
(1) $\frac{m_1}{m_2}$ (2) $\frac{m_2}{m_1}$
(3) $\frac{1}{4}$ (4) $\frac{1}{1}$

15. Given below are two statements:
Statement I: Nuclear force between two nucleons is repulsive for distance between them larger than 0.8 fm.
Statement II: The nuclear mass is always less than the total mass of its constituents.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

16. In Bohr's model of hydrogen atom if P.E. represents potential energy and T.E. represents the total energy of electron, then in going from lower to a higher orbit
(1) P.E. increases but T.E. decreases
(2) Both P.E. and T.E. increases
(3) P.E. decreases but T.E. increases
(4) Both P.E. and T.E. decreases

17. The minimum wavelength in Lyman series is (R is Rydberg constant and c is speed of light in air)
(1) $\frac{1}{R}$ (2) R
(3) $\frac{1}{Rc}$ (4) Rc

18. Choose the correct statement(s).
(1) Pure semiconductors are called intrinsic semiconductors.
(2) Extrinsic semiconductor possesses an overall charge neutrality.
(3) For insulators, energy band gap is 0.2 eV to 3 eV while for semiconductors energy band gap is greater than 3 eV.
(4) Both (1) and (2)

19. Let potential energy of hydrogen atom in ground state to be zero, then the total energy in first excited state will be
(1) 23.8 eV (2) 27.2 eV
(3) 54.4 eV (4) 36.9 eV

20. The minimum energy to ionise an atom is the energy required to
(1) Add one electron to the atom.
(2) Excite the atom from its ground state to its first excited state.
(3) Remove one outermost electron from the atom.
(4) Excite the atom from its ground state to its fourth excited state.

21. For a telescope whose objective has a focal length of 100 cm and eyepiece of focal length 1 cm. The magnifying power of this telescope when eye is in relaxed condition is
(1) 1 (2) 100
(3) 101 (4) 0.01

Space for Rough Work

22. Consider the following statements and choose the correct option.

Statement 1: In intrinsic semiconductor, the number of free electrons n_e is equal to the it is number of holes n_h.

Statement 2: In intrinsic semiconductor, the free electron moves completely independently as conduction electron and give rise to electron current, i_e under an applied electric field.

Statement 3: The total current I in intrinsic semiconductor under the action of an electric field is $I = I_e + I_h$.

(1) Only statement 1 is correct

(2) Only statement 2 is correct

(3) Both statements 1 and 2 are correct but statement 3 is incorrect

(4) All the statements are correct

23. The radius of ^{70}X nucleus is (R_0 = 1.2 fm, $(70)^{1/3} = 4.12$)

(1) 4.94 fm (2) 2.83 fm

(3) 1.53 fm (4) 2.22 fm

24. In the nuclear decay given below

$${}^{A}_{Z}X \longrightarrow {}^{A}_{Z-1}Y \longrightarrow {}^{A-4}_{Z-3}B^{*} \longrightarrow {}^{A-4}_{Z-3}B,$$

the particles emitted in the sequence are

(1) γ, α, β^+ (2) γ, β^-, α

(3) β^+, α, γ (4) α, β^-, γ

25. The electron concentration (n_e) and hole concentration (n_h) in a semiconductor in thermal equilibrium is related with intrinsic carrier concentration (n_i) as

(1) $n_i n_h = n_e^2$ (2) $n_e n_i = n_h^2$

(3) $\dfrac{n_e}{n_h} = n_i^2$ (4) $n_e n_h = n_i^2$

26. The binding energy of deuteron is 'a' MeV and that of Helium nucleus $\left({}^{4}_{2}\text{He}\right)$ is 'b' MeV. If two deuterons are fused to form one ${}^{4}_{2}\text{He}$, then energy released is (in MeV)

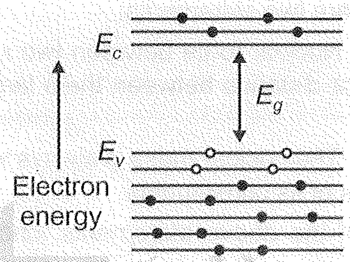
(1) $b - a$

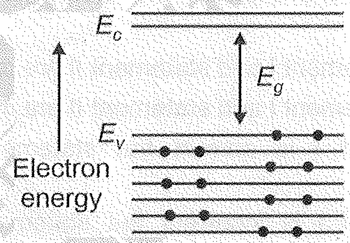
(2) $a - b$

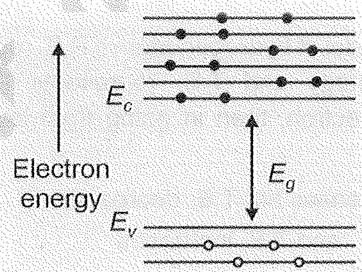
(3) $b - 2a$

(4) $b + 2a$

27. Choose the correct energy band diagram for intrinsic semiconductor at $T > 0$ K (The filled circles (•) represent electrons and empty circles (o) represent holes).

(1) 

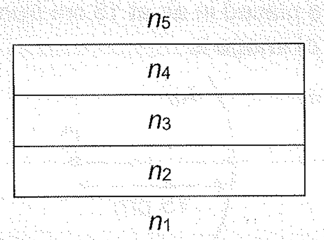
(2) 

(3) 

(4) None of these

Space for Rough Work

28. A ray of light passes through multiple glass slabs of different refractive indices, incident from medium (n_1) and emergent to medium (n_5). If $n_1 = n_5$, then



(1) Incident ray and emergent ray are parallel to each other

(2) Incident ray and emergent ray are perpendicular to each other

(3) Incident ray and emergent ray are never parallel to each other

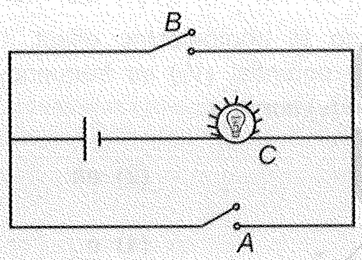
(4) Incident ray and emergent ray are inclined 60° with each other

29. Nuclides with same neutron number N but different atomic number Z, are called

(1) Isotopes (2) Isotones

(3) Isobars (4) Isoelectronic

30. Identify the equivalent logic gate represented by the given circuit.



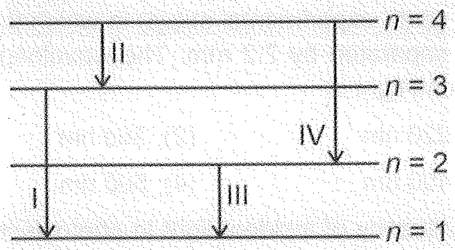
(1) AND gate

(2) NOR gate

(3) OR gate

(4) NAND gate

31. Consider the following energy level diagram for electron in a certain atom. The transition that represents the emission of a photon with the most energy is



(1) I (2) II

(3) III (4) IV

32. A converging beam of rays is incident on a converging lens. Having passed through the lens, the rays intersect at a point 10 cm from the lens on the opposite side. If the lens is removed, the point where the rays meet is 20 cm from the position of lens. The focal length of the lens is

(1) 10 cm

(2) 20 cm

(3) 30 cm

(4) 15 cm

33. Phenomenon explained by wave theory of light is/are

(1) Reflection

(2) Interference

(3) Diffraction

(4) All of these

34. Searle's apparatus is used to determine

(1) Spring constant of a helical spring

(2) Young's modulus of material

(3) Shear modulus of a material

(4) Refractive index of a glass slab

Space for Rough Work

35. In YDSE, laser light of wavelength 630 nm incident on a pair of slits produces an interference pattern in which the fringes are separated by 8.1 mm. A second laser light produces an interference pattern (on same experimental setup) in which the fringes are separated by 7.2 mm. The wavelength of the second light is

(1) 220 nm (2) 560 nm

(3) 400 nm (4) 500 nm

36. If frequency of incident light in photoelectric effect is doubled, then

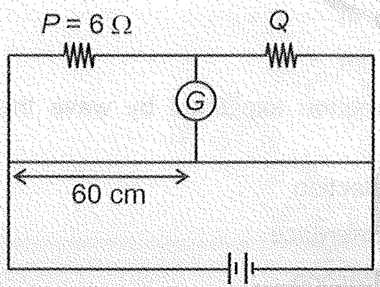
(1) Maximum kinetic energy of photoelectrons is decreased to half.

(2) Stopping potential is reduced to half.

(3) Maximum speed of photoelectron becomes more than $\sqrt{2}$ times.

(4) Frequency of incident light cannot be doubled by any means.

37. For the given meter bridge arrangement if null point is at 60 cm, then the value of Q is



(1) 2 Ω (2) 3 Ω

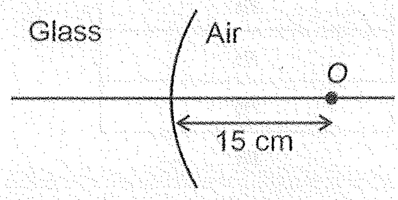
(3) 4 Ω (4) 6 Ω

38. In previous question if 12 Ω is connected in parallel to Q, then shift in null point is nearly

(1) 2.7 cm (2) 5 cm

(3) 6.7 cm (4) 8 cm

39. A convex surface of radius of curvature 30 cm separates glass of refractive index $\frac{3}{2}$ from air. If a point object is placed in air at 15 cm from interface. The position of image is



(1) 18 cm inside air from surface

(2) 20 cm inside glass from surface

(3) 10 cm inside glass from surface

(4) 5 cm inside air from surface

40. The shape of wavefront produce by a large linear source at finite distance is

(1) Spherical (2) Conical

(3) Parabolic (4) Cylindrical

41. If energy required to remove an electron from hydrogen atom from an orbit is 13.6 eV, then the orbital radius of electron in this orbit of hydrogen atom is

(1) 3.5×10^{-11} m (2) 2.5×10^{-11} m

(3) 3.6×10^{-10} m (4) 5.3×10^{-11} m

42. According to photoelectric effect, the slope of stopping potential (V_0) vs frequency of incident radiation (ν) curve is

(1) $\frac{h}{e}$ (2) eh

(3) h (4) e

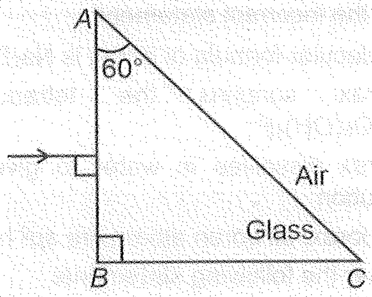
43. The de-Broglie wavelength of an electron revolving in first excited state in hydrogen atom is nearly

(1) 1.11π Å (2) 2.12π Å

(3) 4.22π Å (4) 3.12π Å

Space for Rough Work

44. Light is incident on a face AB of a right-angled glass prism as shown below. If refractive index of the glass is $\sqrt{3}$, then angle of emergence from face AC of the prism is



(1) 30° (2) 45°
(3) 60° (4) Does not exist

45. Consider the following statements:

Statement (A): In forward bias n-side of diode is connected to negative terminal of the battery.

Statement (B): When diode is in forward bias, the width of depletion layer decreases.

(1) Statement A is true but B is false
(2) Statement A is false but B is true
(3) Both Statement A and B are true
(4) Both Statement A and B are false

[CHEMISTRY]

46. Given below are two statements:

Statement I: Lactose on hydrolysis gives α-D-glucose and β-D-galactose

Statement II: Lactose is a reducing sugar.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

47. Consider the following statements about glucose

(a) It is an aldohexose.
(b) It is obtained by hydrolysis of cellulose.
(c) It forms hydrogensulphite addition product with $NaHSO_3$.

The correct statements are

(1) (a) and (b) only
(2) (a) and (c) only
(3) (b) and (c) only
(4) (a), (b) and (c)

48. The non-essential amino acid among the following is

(1) Arginine (2) Tyrosine
(3) Lysine (4) Histidine

49. Given below are two statements:

Statement I: Amino acids are water soluble, high melting solids and behave like salts.

Statement II: In zwitter ionic form, amino acids show amphoteric behaviour.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both Statement I and Statement II are correct
(2) Statement I is correct but Statement II is incorrect
(3) Both Statement I and Statement II are incorrect
(4) Statement I is incorrect but Statement II is correct

50. During denaturation of proteins which structure(s) remain(s) intact?

(1) Tertiary structure only
(2) Primary structure only
(3) Both secondary and tertiary structures
(4) Both primary and secondary structures

Space for Rough Work

51. Given below are two statements:

Statement I: Sugar moiety present in RNA molecule is β-D-glucose.

Statement II: Complete hydrolysis of DNA yields a pentose sugar, nitrogen containing heterocyclic compounds (called bases) and phosphoric acid.

In the light of the above statements, choose the correct answer from the options given below.

(1) Statement I is correct but Statement II is incorrect

(2) Both Statement I and Statement II are incorrect

(3) Both Statement I and Statement II are correct

(4) Statement I is incorrect but Statement II is correct

52. Which of the following is not a fat-soluble vitamin?

(1) Vitamin D (2) Vitamin A

(3) Pyridoxine (4) Vitamin K

53. Number of chiral carbons present in β-D-(+)-Glucopyranose is

(1) 5 (2) 4

(3) 6 (4) 3

54. Consider the following statements:

(a) Amylose is water soluble component which constitutes about 15-20% of starch.

(b) Amylopectin is insoluble in water and constitutes about 80-85% of starch.

(c) In amylopectin, branching occurs by C1-C4 glycosidic linkage.

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

55. Which of the following is not a pyrimidine base?

(1) Cytosine (2) Thymine

(3) Guanine (4) Uracil

56. Which of the following species does not act as a Lewis acid?

(1) B_2H_6 (2) $AlCl_3$

(3) BF_3 (4) NH_3

57. Select the incorrect statement

(1) Molecular formula of borax is $Na_2B_4O_7{\cdot}10H_2O$

(2) Borax contains the tetranuclear units $[B_4O_5(OH)_4]^{2-}$

(3) Borax dissolves in water to give an alkaline solution

(4) In borax, all boron atoms are sp^2 hybridised

58. Consider the following statements

(a) Both white and red phosphorus are soluble in CS_2.

(b) P-P-P bond angle in white phosphorus is 60°.

(c) Red phosphorus is obtained by heating white phosphorus at 573 K in an inert atmosphere for several days.

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

59. In Ostwald's process, nitric acid is prepared by the catalytic oxidation of

(1) N_2 (2) NO

(3) NO_2 (4) NH_3

60. The correct E–H bond dissociation energy in the hydrides of group 15 elements is (E is group 15 element)

(1) $SbH_3 > AsH_3 > NH_3 > PH_3$

(2) $NH_3 > PH_3 > AsH_3 > SbH_3$

(3) $PH_3 > NH_3 > AsH_3 > SbH_3$

(4) $AsH_3 > PH_3 > NH_3 > SbH_3$

61. When copper is treated with conc. H_2SO_4, the oxidation state of sulphur changes from

(1) +6 to 0 (2) +6 to –2

(3) +6 to +4 (4) +6 to +2

Space for Rough Work

62. Consider the following statements about phosphine
(a) It is a colourless gas with rotten fish like smell
(b) Solution of phosphine in water decomposes in presence of light giving red phosphorus and H_2
(c) It gives phosphonium compounds with acids
(d) It is prepared by heating white phosphorus with concentrated NaOH solution in an inert atmosphere of CO_2

The correct statements are
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (d) only (4) (a), (b), (c) and (d)

63. Match List-I with List-II.

	List-I Species Name		List-II Formula
a.	Tear gas	(i)	$ClCH_2CH_2SCH_2CH_2Cl$
b.	Phosgene	(ii)	CCl_3NO_2
c.	Laughing gas	(iii)	$COCl_2$
d.	Mustard gas	(iv)	N_2O

Choose the correct answer from the options given below.
(1) a(i), b(ii), c(iv), d(iii) (2) a(i), b(iii), c(ii), d(iv)
(3) a(ii), b(iii), c(i), d(iv) (4) a(ii), b(iii), c(iv), d(i)

64. When Xe reacts with F_2 in the ratio of 1:5 at 873 K, 7 bar pressure, the product formed is
(1) XeF_2 (2) XeF_4
(3) XeF_6 (4) XeO_2F_2

65. Match List I with List II

	List I Formula		List II Physical state and colour
a.	ClF_3	(i)	Yellow green liquid
b.	BrF_3	(ii)	Orange solid
c.	IF_3	(iii)	Colourless gas
d.	ICl_3	(iv)	Yellow powder

Choose the correct answer from the options given below.
(1) a(iii), b(i), c(iv), d(ii) (2) a(iv), b(iii), c(i), d(ii)
(3) a(iii), b(i), c(ii), d(iv) (4) a(iv), b(i), c(iii), d(ii)

66. Which among the following pair is the example of fibrous proteins?
(1) Albumin and insulin (2) Keratin and myosin
(3) Albumin and myosin (4) Keratin and insulin

67. Riboflavin is also known as
(1) Vitamin B_6 (2) Vitamin B_1
(3) Vitamin B_2 (4) Vitamin C

68. The main product obtained, when glucose reacts with nitric acid
(1) Gluconic acid (2) Saccharic acid
(3) Citric acid (4) Lactic acid

69. On addition of β-naphthol to diazonium salt, colour obtained is
(1) Orange (2) Yellow
(3) Blue (4) Scarlet red

70. $\text{Salt} + AgNO_3 \longrightarrow \underset{\text{Yellow ppt}}{(A)}$

"A" is almost insoluble in concentrated NH_3 solution. The anion in the salt is
(1) Cl^- (2) CO_3^{2-}
(3) S^{2-} (4) I^-

71. When H_2S is passed through an ammoniacal salt solution(X), a white ppt is obtained. (X) can be
(1) Cobalt salt (2) Nickel salt
(3) Manganese salt (4) Zinc salt

72. During the identification of basic radical in a salt, when its small amount was treated with a few drops of sodium hydroxide then a gas 'X' with characteristic odour is evolved. Gas 'X' react with Nessler's reagent to form a brown coloured precipitate. The basic radical is
(1) NH_4^+ (2) Ba^{2+}
(3) Al^{3+} (4) Fe^{3+}

Space for Rough Work

73. On adding dilute H_2SO_4 to the solid salt, colourless and odourless gas evolved with brisk effervescence which turns lime water milky. The anion present in the salt is

(1) SO_4^{2-} (2) CO_3^{2-}

(3) I^- (4) Br^-

74. Which reagent can be used to identify Ni^{2+} ion in solution?

(1) Grignard reagent

(2) Dimethyl glyoxime

(3) Potassium ferrocyanide

(4) Diphenyl benzidine

75. In qualitative analysis when H_2S gas is passed through an aqueous solution of salt acidified with dilute HCl, a black ppt. is obtained. On boiling the precipitate with dilute HNO_3 it forms a solution of blue colour. Addition of excess of aqueous solution of NH_3 to this solution gives

(1) Deep blue ppt of $Cu(OH)_2$

(2) Deep blue solution of $[Cu(NH_3)_4]^{2+}$

(3) Deep blue solution of $Cu(NO_3)_2$

(4) Deep blue solution of $Cu(OH)_2 \cdot Cu(NO_3)_2$

76. Range of size of colloidal particle is

(1) 0.1 nm to 1 nm (2) 1 nm to 1000 nm

(3) 100 nm to 1000 nm (4) 10^{-2} nm to 1 nm

77. A glassy bead formed by heating borax on a platinum wire loop is:

(1) Sodium tetraborate only

(2) Sodium metaborate only

(3) Sodium metaborate and boric anhydride

(4) Boric anhydride and sodium tetraborate

78. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In vapour state, S_2 molecule exhibits paramagnetism.

Reason (R) : S_2 molecule has 1 unpaired electron in the antibonding π^* orbitals.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

79. 1.96 g of ferrous ammonium sulphate (Mohr's salt) reacts completely with 20 mL $\frac{N}{10}$ $KMnO_4$ solution. The percentage purity of the sample is (At wt. of Fe = 56 u)

(1) 50 (2) 78.4

(3) 40 (4) 29.2

80. Which reaction is not feasible?

(1) $2KI + Br_2 \rightarrow 2KBr + I_2$

(2) $2KBr + I_2 \rightarrow 2KI + Br_2$

(3) $2KBr + Cl_2 \rightarrow 2KCl + Br_2$

(4) $2H_2O + 2F_2 \rightarrow 4HF + O_2$

81. Match List-I with List-II.

	List-I (Element)		**List-II (Flame colour observed by naked eye)**
a.	Copper	(i)	Apple green
b.	Barium	(ii)	Brick red
c.	Strontium	(iii)	Crimson red
d.	Calcium	(iv)	Green with blue centre

Choose the correct answer from the options given below.

(1) a(iv), b(i), c(iii), d(ii) (2) a(iv), b(i), c(ii), d(iii)

(3) a(i), b(iv), c(iii), d(ii) (4) a(ii), b(iii), c(i), d(iv)

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82. Given below are two statements:

Statement I: $PbCrO_4$ is highly soluble in water

Statement II: $PbCrO_4$ is soluble in hot sodium hydroxide solution.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

83. Match List-I with List-II.

	List-I Cation group number		**List-II Group reagent**
a.	Group-I	(i)	H_2S gas in presence of dil.HCl
b.	Group-II	(ii)	H_2S in presence of NH_4OH
c.	Group-III	(iii)	Dilute HCl
d.	Group-IV	(iv)	NH_4OH in presence of NH_4Cl

Choose the correct answer from the options given below.

(1) a(iv), b(ii), c(iii), d(i) (2) a(i), b(iii), c(iv), d(ii)

(3) a(iii), b(i), c(ii), d(iv) (4) a(iii), b(i), c(iv), d(ii)

84. The colour of methyl orange in basic medium is

(1) Red (2) Green

(3) Orange (4) Yellow

85. Cation that gives white residue with the odour of garlic in charcoal cavity test is

(1) Pb^{2+} (2) As^{3+}

(3) Cd^{2+} (4) Zn^{2+}

86. Colour of complex formed by iodine with starch is

(1) Yellow (2) Blue

(3) Pink (4) Green

87. Which among the following will not give positive Tollens' test?

(1) Maltose (2) Lactose

(3) Sucrose (4) Fructose

88. Maltose is composed of

(1) β-D-galactose and β-D-glucose

(2) α-D-glucose and α-D-glucose

(3) β-D-glucose and α-D-glucose

(4) α-D-glucose and β-D-fructose

89. The major product obtained on prolonged heating of glucose with HI is

(1) n-Hexane (2) Gluconic acid

(3) Saccharic acid (4) Lactic acid

90. Optically inactive amino acid among the following is

(1) Valine (2) Lysine

(3) Glycine (4) Alanine

[BOTANY]

91. Match Column-I with Column-II and select the **correct** option.

	Column-I		**Column-II**
a.	Aravalli hills	(i)	Madhya Pradesh
b.	Khasi and Jaintia hills	(ii)	Karnataka and Maharashtra
c.	Western Ghat	(iii)	Meghalaya
d.	Sarguja, Chanda and Bastar	(iv)	Rajasthan

(1) a(iii), b(ii), c(i), d(iv) (2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(iv), c(i), d(iii) (4) a(i), b(iv), c(ii), d(iii)

92. All are the characteristics of anthropogenic ecosystem, **except**

(1) They do not possess self-regulatory mechanism

(2) They have little diversity

(3) They have simple food chain

(4) They have low productivity

Space for Rough Work

93. Read the following statements and select the **correct** option.

Statement A: Producers are also known as transducers because they convert solar energy into chemical energy.

Statement B: Key industry animals are those which convert plant matter into animal matter

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Both the statements A and B are incorrect

94. Select the **odd** one out w.r.t. key functional aspects of the ecosystem.

(1) Productivity (2) Decomposition
(3) Stratification (4) Energy flow

95. Read the following statements (A to D) and select the option for **correct** ones.

(A) NPP is the rate of organic matter buildup by producers in excess of respiratory utilization per unit time and area.
(B) Productivity is the rate of biomass production.
(C) GPP is the rate of production of organic matter by producers during photosynthesis.
(D) Vertical stratification in forests is less diverse than that of desert ecosystem.

(1) A, B and D only (2) A and C only
(3) B and D only (4) A, B and C

96. Select **odd** one out w.r.t. anthropogenic ecosystem.

(1) Estuaries (2) Crop field
(3) Aquarium (4) Garden

97. Select the **mismatched** pair w.r.t. biodiversity in Amazonian rain forest.

	Taxa		**Number of species**
(1)	Mammals	–	427
(2)	Birds	–	1300
(3)	Reptiles	–	378
(4)	Amphibians	–	3000

98. Primary productivity is expressed in terms of

(1) $(\text{kcal m}^{-2})\text{yr}^{-2}$ (2) $(\text{gm m}^{-1})\text{yr}^{-2}$
(3) $(\text{kcal m}^{-2})\text{yr}^{-1}$ (4) $\frac{\text{kcal}}{\text{m}^2}$

99. Read the following statements and identify them as **True (T)** or **False (F)**.

(a) Temperature is the most ecologically relevant environmental factor.
(b) Mango trees are tropical and subtropical plants, and are not found in temperate countries.
(c) Global warming will push tropics into temperate areas and temperate areas towards pole.
(d) In *Opuntia*, leaves are modified into spines to minimize the water loss.

	(a)	(b)	(c)	(d)
(1)	T	F	F	F
(2)	F	T	T	F
(3)	T	T	T	T
(4)	F	F	T	F

100. Organisms which are restricted to a narrow range of salinities are called

(1) Stenothermal (2) Euryhaline
(3) Eurythermal (4) Stenohaline

101. Mammals from colder climates generally have shorter ears and limbs to minimise heat loss. This is called

(1) Allen's rule (2) Jordan's rule
(3) Bergmann's rule (4) Rensch's rule

102. Number of births during a given period in the population that are added to the initial density refers to

(1) Natality (2) Mortality
(3) Immigration (4) Emigration

103. Select **odd** one out w.r.t. population attributes.

(1) Sex ratio (2) Per capita deaths
(3) Per capita births (4) Death

Space for Rough Work

104. In which of the following population interactions both the species show detrimental effect to each other?
(1) Predation (2) Amensalism
(3) Commensalism (4) Competition

105. Amount of living material present in different trophic levels at a given time is known as
(1) Standing crop (2) Standing state
(3) Standing quality (4) Dry weight

106. Identify the **incorrect** statement w.r.t. different hierarchial level of biodiversity.
(1) Genetic diversity enables a population to adapt to its environment and the changes occurring in the environment.
(2) Maximum taxonomic diversity occurs where species of taxonomically different groups occur in almost equal abundance
(3) Species diversity is the product of species richness and species unevenness
(4) Ecosystem diversity is high in India because of the occurrence of a large number of ecosystems

107. Which of the following statements is **incorrect** w.r.t. age pyramid for human population?
(1) It is a graphic representation of proportion of various age groups of a population
(2) Population is said to be stable if there is no increase or decrease in its size due to growth rate being almost zero
(3) In urn-shaped age pyramid population has a very high proportion of pre-reproductive individuals than reproductive individuals
(4) The age pyramid is bell-shaped when pre-reproductive individuals are only marginally more than the reproductive individuals

108. The **correct** equation of species-area relationship on a logarithmic scale is
(1) log S = log C + A log Z
(2) log Z = log S + A log C
(3) log S = log C + Z log A
(4) log Z = log C + S log A

109. Read the following statements (A - D).
(A) The hotspots are the richest and the most threatened reservoirs of plants and animals.
(B) Collection of forest products, harvesting of timber, tilling of land are not allowed in wildlife sanctuaries.
(C) The *in-situ* conservation strategies emphasise protection of whole ecosystem, therefore its biodiversity at all levels is protected.
(D) Drugs, industrial product and aesthetic pleasure are categorised in narrowly utilitarian services of ecosystem.

In the light of the above statements, select the **correct** statement from the options given below:
(1) Only (A), (B) and (D) (2) Only (A) and (D)
(3) Only (B) and (C) (4) Only (A) and (C)

110. The most important cause driving animals and plants to extinction is
(1) Over-exploitation
(2) Alien species invasions
(3) Co-extinctions
(4) Habitat loss and fragmentation

111. The world summit on sustainable development was held in the year __A__ in __B__.

	A	B
(1)	2004	South America
(2)	2002	South Africa
(3)	1992	South Asia
(4)	1994	Australia

112. Which of the following organisms has become recently extinct from Russia?
(1) Thylacine (2) Quagga
(3) Dodo (4) Steller's sea cow

113. Read the following statements of Assertion (A) and Reason (R) and select the **correct** option.
Assertion (A): Plants can avoid being eaten by their predators like phytophagous insects.

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Reason (R): Plants have evolved an astonishing variety of morphological and chemical defence against herbivores.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

114. A stage of suspended development found in many zooplankton species is called

(1) Migration (2) Aestivation

(3) Diapause (4) Hibernation

115. Identify the **incorrect** statements w.r.t. adaptation shown by parasites.

(1) Presence of adhesive organs

(2) Loss of digestive system

(3) Loss of unnecessary sense organs

(4) Low reproductive capacity

116. Which of the following statements is **not** true?

(1) Species composition is identification and enumeration of plant and animal species of an ecosystem.

(2) Vertical distribution of different species occupying different levels is also considered as functional component of an ecosystem.

(3) Detritus rich in lignin and tannins undergoes slow decomposition.

(4) Availability of nutrients and soil moisture affect the magnitude of primary productivity.

117. Match Column-I with Column-II and select the **correct** option.

	Column-I		Column-II
a.	Mutualism	(i)	Ticks on dogs
b.	Commensalism	(ii)	Prickly pear cactus and moth
c.	Predation	(iii)	Orchids growing on branch of mango tree
d.	Parasitism	(iv)	Mycorrhiza

(1) a(iv), b(iii), c(i), d(ii) (2) a(iii), b(iv), c(i), d(ii)

(3) a(iv), b(iii), c(ii), d(i) (4) a(ii), b(i), c(iv), d(iii)

118. Identify the **incorrect** statement w.r.t. limitations of ecological pyramid

(1) It does not take into account the same species belonging to two or more trophic levels

(2) It assumes a simple food chain

(3) Except for saprophytes; decomposers, microbes and detritivores are not given any place in ecological pyramid.

(4) It does not accommodate a food web

119. The process of decomposition of detritus to form dark-coloured, amorphous, more or less decomposed organic matter rich in cellulose, lignin is called

(1) Leaching (2) Humification

(3) Mineralisation (4) Fragmentation

120. Read the following statements (A–D)

(A) Annual NPP of whole biosphere is approximately 170 billion tons (dry weight) of organic matter.

(B) Pyramid of number is upright in grassland ecosystem.

(C) Pyramid of energy is always upright except pond ecosystem.

(D) Pyramid of number can be spindle shaped in tree ecosystem.

(E) Pyramid of biomass is generally inverted in sea ecosystem.

The **correct** ones are:

(1) All except E (2) A, C and E only

(3) All except C (4) D and E only

121. If in a food chain, energy available to herbivores is 5000 J then the amount of energy which gets stored and fixed in tertiary consumer for next trophic level will maximally be

(1) 50 J (2) 500 J

(3) 5 J (4) 0.5 J

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122. The region of biosphere reserve which is managed to accommodate a greater research and educational activities is
(1) Core zone (2) Buffer zone
(3) Transition zone (4) Natural zone

123. Most important climatic factors that regulate the rate of decomposition are
(1) Temperature and soil moisture
(2) Soil pH and aeration
(3) Aeration and temperature
(4) Moisture and soil pH

124. Identify the possible link 'A' in the following food chain.
Grass $\rightarrow$ Grasshopper $\rightarrow$ Frog $\rightarrow$ A $\rightarrow$ Eagle
(1) Zooplankton (2) Rabbit
(3) Snake (4) Sparrow

125. Match Column-I with Column-II and select the **correct** option.

	Column-I		Column-II
a.	Edward Wilson	(i)	Global species diversity at about 7 million
b.	Alexander von Humboldt	(ii)	Rivet popper hypothesis
c.	Paul Ehrlich	(iii)	Species-Area relationship
d.	Robert May	(iv)	Popularised the term biodiversity

(1) a(iv), b(iii), c(ii), d(i) (2) a(iv), b(iii), c(i), d(ii)
(3) a(iii), b(iv), c(ii), d(i) (4) a(ii), b(i), c(iv), d(iii)

126. Which one of the following animals may occupy more than one trophic level in the same ecosystem at the same time?
(1) Butterfly caterpillar (2) Tadpole
(3) Goat (4) Sparrow

127. Identify the **correct** statement w.r.t. energy flow in an ecosystem.
(1) Energy flow in an ecosystem is always multidirectional
(2) Plants capture 2-10% of incident solar radiation for photosynthesis
(3) Energy remains trapped in any organism permanently
(4) Energy flow follows law of thermodynamics

128. Nile perch, a large predator fish introduced into Lake Victoria of East Africa is an example of
(1) Alien species invasion
(2) Over-exploitation
(3) Co-extinctions
(4) Habitat loss and fragmentation

129. Read the following statements (A–D)
(A) Kangaroo rat, can concentrate its urine so that minimal volume of water is used to remove excretory product.
(B) Desert lizards manage the body temperature by behavioural means.
(C) Archaebacteria sustain their life under extremes of temperature due to branched chain lipids in cell membrane.
(D) Bear escapes in time during winter by a process called aestivation.
In the light of the above statements, choose the options for **correct** ones.
(1) (A) and (B) only (2) (B) and (D) only
(3) (A), (B) and (C) (4) (A), (C) and (D)

130. A biologist studied the population of rats in a barn. He found that during a period of time the number of births was 350, number of deaths was 340, number of immigrants was 20 and number of emigrants was 30. The net increase in population during that period is
(1) 20 (2) 30
(3) 55 (4) Zero

131. The integral form of exponential growth equation of a population is
(1) $N_0 = N_t e^{rt}$ (2) $N_t = N_0 e^{rt}$
(3) $N_t - N_0 = e^{rt}$ (4) $N_t + N_0 = \frac{1}{e^{rt}}$

Space for Rough Work

132. Which of the following statements is **not** correct w.r.t. decomposers?
   (1) They are saprophytic microorganisms deriving food material from organic matter present in dead organism
   (2) They are natural scavengers as they reduce organic remains of earth
   (3) They replenish the soil naturally with minerals that are essential for growth of plants
   (4) They secrete digestive enzymes which form complex organic substances from simpler ones

133. Consider the following data:

Trophic level		**No. of individual**
Primary producer	=	58460
Tertiary consumer	=	80
Primary consumer	=	5846
Secondary consumer	=	485

In the light of the above data, select the food chain which are possible and choose the option accordingly.
   (A) Aquatic plants → Small fish → Large fish → Fish eating bird
   (B) Grass → Grasshopper → Frog → Snake
   (C) Large tree → Insects → Small bird → Hawk
   (D) Grasses → Rabbit → Snake → Eagle
   (1) Only (A) and (C) (2) Only (B) and (C)
   (3) (A), (B) and (D) (4) Only (A) and (D)

134. Read the following statements and select the **correct** option.
   Statement A: Biological pest control methods used in agriculture are based on the ability of predator to regulate prey population.
   Statement B: Predators help in maintaining species diversity in a community by reducing the intensity of competition among competing prey.
   (1) Only statement A is correct
   (2) Only statement B is correct
   (3) Both the statements A and B are correct
   (4) Both the statements A and B are incorrect

135. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
   Assertion (A): In general, communities with more species tend to be more stable than those with less species.
   Reason (R): A stable community show too much variation in productivity from year to year.
   (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
   (2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
   (3) (A) is true but (R) is false
   (4) Both (A) and (R) are false

[ZOOLOGY]

136. Which of the following scientists developed a method of utilising small ringlets of DNA as the vector?
   (1) Herbert Boyer
   (2) Stanley Cohen
   (3) S. L. Miller
   (4) J.D. Watson

137. To introduce genetically modified cow that produce protein enriched milk in Indian markets, permission from which of the following organisations will be necessary?
   (1) NACO (2) GEAC
   (3) WHO (4) EFB

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138. Recognition site of which of the following restriction enzymes is found within *amp*^{R} gene of pBR322?
(1) *Eco* RI (2) *Pvu* II
(3) *Pvu* I (4) *Sal* I

139. An enzyme belonging to ______ plays a vital role in the formation of rDNA as it helps in linking the alien DNA and cloning vector and is also called ______.
Select the correct option to fill in the blanks respectively.
(1) Class VI; molecular scissors
(2) Class IV; molecular glue
(3) Class VI; molecular glue
(4) Class IV; molecular scissors

140. In the restriction enzyme *Bam*HI, 'H' refers to the
(1) Genus of the source organism
(2) Species of the source eukaryotic cell
(3) Name of the strain
(4) Order of isolation

141. Complete the analogy by selecting the **correct** option.
Formation of recombinant protein : Biosynthetic stage : : Separation and purification of protein : ______.
(1) Downstream processing
(2) Upstream processing
(3) Transformation process
(4) Gel electrophoresis

142. "A foreign DNA is inserted at the *EcoRI* sequence of pBR322 and introduced in *E.coli*".
What will be the **correct** implication in recombinant *E.coli* w.r.t. the above statement?
(1) *E.coli* will show sensitivity to both tetracycline and ampicillin
(2) *E.coli* will show resistance against both tetracycline and ampicillin
(3) *E.coli* will show sensitivity to tetracycline and resistance against ampicillin
(4) *E.coli* will show resistance against tetracycline only

143. Today transgenic models exist for human diseases such as
(a) Cancer (b) Alzheimer's disease
(c) Cystic fibrosis
Choose the correct option.
(1) Only (b) and (c) (2) Only (a)
(3) (a), (b) and (c) (4) Only (c)

144. All of the following serve the purpose of early diagnosis for HIV positive status in a man who received infected blood few months back, **except**
(1) Recombinant DNA technology
(2) Polymerase Chain Reaction
(3) Enzyme Linked Immuno-sorbent Assay
(4) Urine analysis

145. Choose the **incorrect** option w.r.t. Green revolution.
(1) Involved the use of only bio-pesticides
(2) Succeeded in tripling the food supply
(3) Increased yields was partially due to the use of improved crop varieties
(4) The increased yield was not enough to feed the growing population

146. In 1990s, an American company got patent rights on a crop derived from Indian semi-dwarf varieties through US Patent and Trademark Office by claiming it as an invention. Similar attempts have been made to patent the use of product of traditional herbal medicine such as
(1) Wheat (2) Tobacco
(3) Neem (4) Bt soyabean

147. Choose the **correct** statement.
(1) Insulin is synthesised in mature form by joining three peptide chains namely chain A, B and C.
(2) Earlier insulin was extracted from gall bladder of the live cattle and pigs.
(3) In 1988, an American company prepared three polypeptides corresponding to namely chain A, B and C chains and introduced them in plasmids.
(4) Chains A and B were produced separately in *E.coli*, extracted and combined by creating two interchain disulfide bonds to form recombinant human insulin.

Space for Rough Work

148. **Assertion (A) :** Micro-injection is a direct method of gene transfer.
Reason (R) : It involves use of disarmed pathogen vectors for transfer of rDNA into the host.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false.

149. Select the **incorrect** option w.r.t. PCR.
(1) During the annealing stage, small oligonucleotides are added at 3'OH of the DNA template
(2) A thermostable enzyme known as *Taq* polymerase is used in this process
(3) Nucleotides such as guanosine and cytosine are added to extend the primer
(4) Can be utilised for diagnostic tests

150. A foreign DNA is inserted within the gene sequence of enzyme β-galactosidase of a plasmid and introduced it in a bacterial host, now in the presence of chromogenic substrate the
(1) Recombinants will produce blue coloured colonies
(2) Recombinants will produce white coloured colonies
(3) Non-recombinants will produce white coloured colonies
(4) Non-transformants will produce blue coloured colonies

151. Select the **correct** match.
(1) Transposons – Enzyme
(2) Agarose – Selectable marker
(3) Lysozyme – Vitamin
(4) Chloramphenicol – Antibiotic

152. During isolation of genetic material, the purified DNA ultimately precipitates out after the addition of
(1) Chilled ethanol (2) Chitinase
(3) Ribonuclease (4) DNA ligase

153. The recombinant DNA can be forced into competent host cells by incubating the cells with recombinant DNA on ice, followed by placing them briefly at
(1) 94°C (2) 80°C
(3) 37°C (4) 42°C

154. In a culturing method named 'X', fresh medium is added from one side while the used medium is removed from another and a large biomass w.r.t. desired protein is obtained by keeping the cells in their
(1) Log phase (2) Lag phase
(3) Stationary phase (4) Decline phase

155. Select the **correct** match.
(1) 1986 – *Hin*d II was isolated and characterised
(2) 1972 – Construction of first rDNA
(3) 1990 – First transgenic cow produced human protein-enriched milk
(4) 1997 – First clinical gene therapy

156. Enzyme isolated from *Escherichia coli*, restricted the growth of bacteriophage by
(1) Adding the methyl group to bacteriophage's DNA
(2) Cutting the DNA of bacterium
(3) Cutting the DNA of bacteriophage
(4) Removing the methyl group from bacterial DNA

157. Select the **incorrect** statement w.r.t. restriction endonuclease.
(1) They functions by inspecting the length of a DNA sequence.
(2) They are naturally found in mammalian cells and cut the double helix of DNA from terminal ends.
(3) They cuts each of two strands of the double helix at specific points.
(4) They are utilised in genetic engineering to form recombinant DNA.

Space for Rough Work

158. To construct a rDNA, the plasmid with *Bam*HI and *Hind*II restriction sites and alien DNA with *Hind*II restriction site were joined together after cutting
   (1) Both the plasmid and alien DNA by *Bam*HI at its specific sequence
   (2) Both the plasmid and alien DNA by *Bam*HI and *Eco*RI
   (3) The plasmid by *Bam*HI and alien DNA by *Hind*II
   (4) Both the plasmid and alien DNA by *Hind*II at its specific sequence

159. Choose the **correct** option to complete the given analogy.
   cryIAc : Cotton bollworms : : *cryIAb* : ______
   (1) Flies
   (2) Corn borers
   (3) Mosquitoes
   (4) Beetles

160. A product obtained from genetically modified organism that is used to treat a chronic disorder in which respiratory surface is decreased, is
   (1) Insulin
   (2) α-1-antitrypsin
   (3) α- lactalbumin
   (4) Insulin-like growth factor

161. Read the given statements and choose the correct option.
   Statement A: During autoradiography, a ssDNA tagged with a radioactive molecule is allowed to hybridise to its complementary DNA in a clone of cells.
   Statement B: ELISA is based on antigen-antibody interaction.
   (1) Both the statements are incorrect
   (2) Both the statements are correct
   (3) Only statement A is correct
   (4) Only statement B is correct

162. Select the **incorrect** statement w.r.t. transgenic animals.
   (1) 95 per cent of all the existing transgenic animals are mice.
   (2) Transgenic monkeys are being developed for safety testing of vaccines and can replace mice very soon.
   (3) They are made to carry genes which make them more susceptible to toxic substances.
   (4) They are utilised to form biological products for the treatment of diseases like phenylketonuria.

163. **Assertion (A):** A bacterial pathogen such as *Meloidogyne* is able to cause tumor in several dicot plants.
   Reason (R): The pathogen delivers a piece of DNA known as T-DNA and transforms tumor into normal cells in plants.
   In the light of above statements, select the **correct** option.
   (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
   (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
   (3) (A) is true but (R) is false
   (4) Both (A) and (R) are false

164. What could be the permanent cure for a genetic disease caused due to the deficiency of adenosine deaminase?
   (1) Gene therapy after birth
   (2) Gene therapy during early embryonic stage
   (3) Bone marrow transplantation at later stage of life
   (4) Enzyme replacement therapy

165. A toxin protein released by some strains of *Bacillus thuringiensis* binds to the surface of midgut epithelial cells and create pores if ingested by
   (1) *Meloidogyne incognita*
   (2) *Agrobacterium tumefaciens*
   (3) Coleopterans
   (4) Retrovirus

Space for Rough Work

166. Match the Column I with Column II and choose the **correct** option.

	Column I		**Column II**
a.	GM plants	(i)	Genetically engineered lymphocytes
b.	ADA deficiency	(ii)	Fuels and pharmaceuticals
c.	Mature insulin	(iii)	Intrachain disulfide bond present in B peptide
		(iv)	Intrachain disulphide bond present in A peptide

(1) a(ii), b(i), c(iv)
(2) a(i), b(iii), c(iv)
(3) a(ii), b(i), c(iii)
(4) a(iii), b(ii), c(i)

167. A novel strategy which involves silencing of specific mRNA to prevent translation is due to complementary

(1) ssRNA (2) dsRNA
(3) ssDNA (4) dsDNA

168. Select the **correct** match.

(1)	Lepidopterans	–	Armyworms
(2)	Lepidopterans	–	Beetles
(3)	Dipterans	–	Tobacco budworms
(4)	Coleopterans	–	Flies

169. All of the statements given below are correct w.r.t. gel electrophoresis, **except**

(1) Smallest DNA fragment is present near anode.
(2) DNA fragments separate due to sieving effect provided by gel.
(3) Separated DNA fragments are extracted from the gel place through a process called spooling.
(4) DNA fragments move towards the positive electrode due to their negative charge.

170. Select the **correct** match.

(1) Stirred tank bioreactor – Flat bladed impeller with foam breaker
(2) Recombinant protein – Protein encoding gene in homologous host
(3) Chitinase – For digestion of DNA of fungal cell
(4) First recombinant DNA – Native plasmid of *Escherichia coli* was used

171. Which of the following can assist in identifying the transformed cells but is absent in pBR322?

(1) *rop* gene (2) β-galactosidase
(3) *Bam*HI (4) *Pvu*I

172. A method of gene transfer which involves bombardment of high velocity microparticles of gold coated with DNA is mainly used for

(1) *Meloidogyne* (2) *Rattus*
(3) Plants (4) Nematodes

173. Select the **odd** one w.r.t. genetically modified plants.

(1) Are more tolerant to salt and heat
(2) Decreased reliance on chemical pesticides
(3) Enhanced nutritional value of food, *e.g.*, golden rice *i.e.*, Vitamin 'C' enriched rice.
(4) Helped to reduce post-harvest losses by increasing the shelf life

174. To recover healthy plants after a viral infection which part of plant can be utilised to grow *in-vitro* virus free plants?

(1) Apical meristem (2) Root
(3) Protoplast (4) Stem

175. How many of the following mentioned in the box below is/are genetically modified?

Pomato, Retrovirus, *Meloidogyne incognita*, Rosie, *Bacillus thuringiensis*

Choose the **correct** option.

(1) Five (2) Four
(3) Three (4) One

Space for Rough Work

176. Arrange the following steps involved in rDNA technology in correct order.
(a) Isolation and separation of the DNA fragment.
(b) Cutting of DNA at specific locations.
(c) Isolation of the genetic material.
(d) Ligation of DNA fragment into vector.
Choose the **correct** option.
(1) $a \rightarrow c \rightarrow b \rightarrow d$
(2) $c \rightarrow b \rightarrow a \rightarrow d$
(3) $b \rightarrow a \rightarrow c \rightarrow d$
(4) $d \rightarrow a \rightarrow b \rightarrow c$

177. Select the **correct** option.
(1) Cloning vector should have very few preferably single recognition site for the commonly used restriction enzymes
(2) Cloning vector should have more than one restriction sites for a restriction enzyme
(3) Normal *E.coli* cells carry natural resistance against ampicillin
(4) *rop* genes are exclusively responsible for controlling the copy numbers of linked DNA

178. All of the following are applications of biotechnology, **except**
(1) Bioremediation
(2) Traditional hybridisation
(3) Processed food
(4) Waste treatment

179. Maintenance of sterile ambience to enable the growth of only desired microbes / eukaryotic cells in large quantities for manufacturing of products is called
(1) Genetic engineering
(2) Bioprocess engineering
(3) Micropropagation
(4) Tissue culture

180. **Assertion (A):** Specific concentration of divalent cations like Ca^{2+} increase the efficiency with which DNA enters the bacterium.
Reason (R): Divalent cations attract positively charged DNA fragments and increase the competency of bacterial cell.
In the light of above statements, choose the **correct** option.
(1) (A) is true and (R) is false.
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) Both (A) and (R) are false.
(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

❑ ❑ ❑

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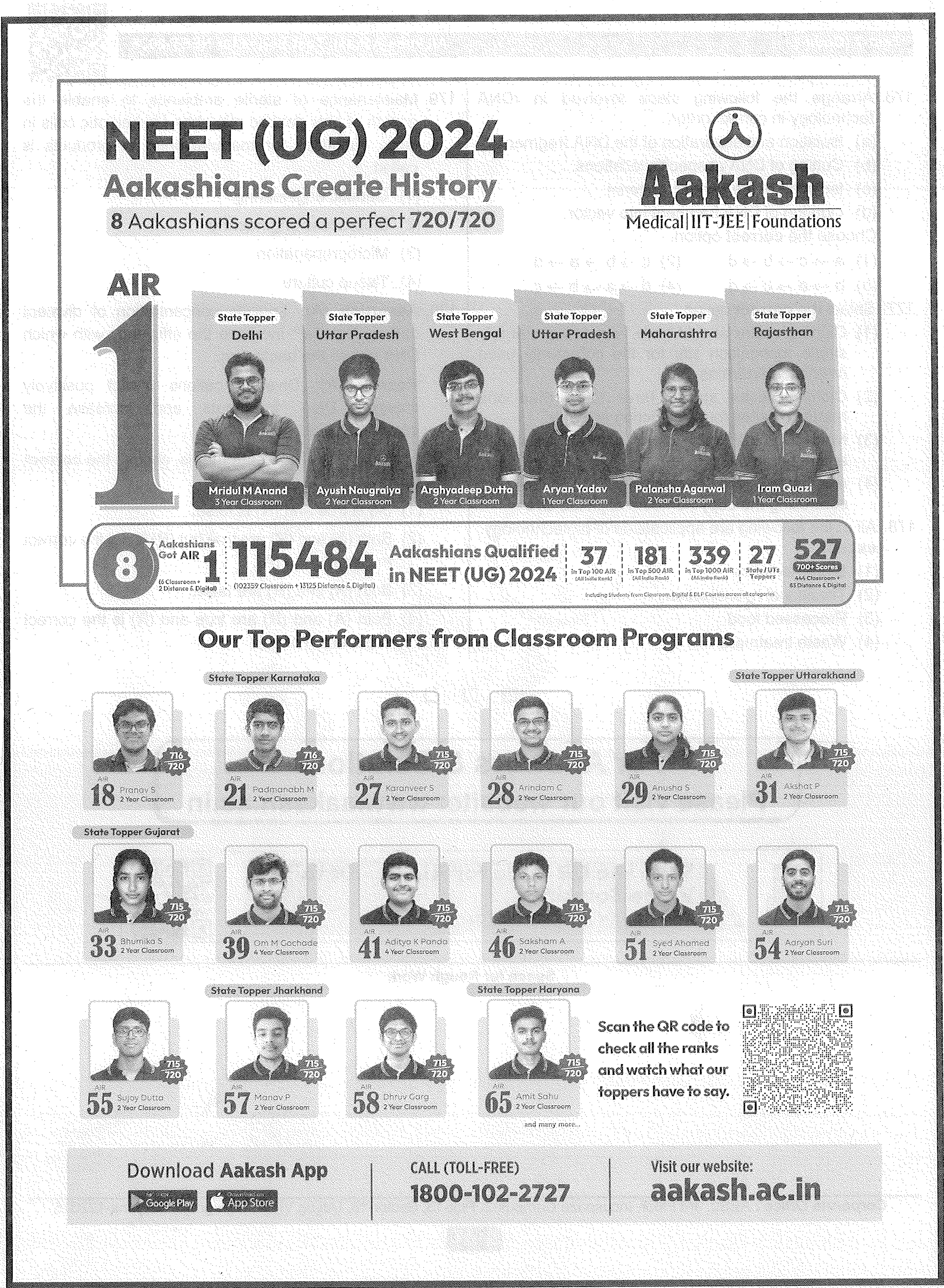
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